2.5 The Derivative of the Trigonometric Functions

Name _____ Period ____ Date _____

AP Calculus 2.5 · Student sheet

I can…

- find the derivatives of sine and cosine.
- find the derivatives of tangent, secant, cotangent, and cosecant.
- use trig derivatives with the Product and Quotient Rules and to find tangent lines.

Vocabulary · fill in the missing word

Derivatives of sine and cosine — $\frac{d}{dx}\sin x =$ _____ $= -\sin x$

Derivatives of the other trig functions — _____

Part 1 · Slopes of sine

Estimate the slope of $y = \sin x$.

x	0	π/2	π	3π/2	2π
Slope of sin x					
cos x					

Part 2 · Notes and examples (fill in each note, then work the example)

Sine and cosine: $\frac{d}{dx}\sin x = \cos x$ and $\frac{d}{dx}\cos x =$ _____.

Example 1

$f(x) = 3\sin x - 2\cos x$. Find $f'(x)$ and $f'(\frac{\pi}{2})$.

Example 2

$f(x) = x\sin x$. Find $f'(x)$ and $f'(\frac{\pi}{2})$.

Tangent and secant: $\frac{d}{dx}\tan x = \sec^2 x$ and $\frac{d}{dx}\sec x =$ _____.

Example 3

$f(x) = \tan x$. Find $f'(\frac{\pi}{3})$, and the tangent line at $(0, 0)$.

Cotangent and cosecant: $\frac{d}{dx}\cot x = -\csc^2 x$ and $\frac{d}{dx}\csc x =$ _____.

Example 4

Use the Quotient Rule to show that $\frac{d}{dx}\csc x = -\csc x \cot x$.

Example 5

Find the limit as $h \to 0$ of $\dfrac{\cos\left(\frac{\pi}{3}+h\right)-\cos\left(\frac{\pi}{3}\right)}{h}$.

Part 3 · Partner practice

Partner 1

$f(x) = x \cos x$. Find $f'(\pi)$.

Partner 2

(AP® Practice) If $y = x \sin x$, then $\frac{dy}{dx} =$

(A) $x \cos x + \sin x$ (B) $x \cos x - \sin x$ (C) $\cos x + \sin x$ (D) $(x + 1)\cos x$